

POLS 641

Introductory Political Analysis

Class 6: Hypothesis Testing

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Warm-Up

In your notes write out your best guess for final project topic

① Simple Arrows: $X \rightarrow Y$

② Using Core Model:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

③ In a sentence:

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- 3 In a sentence:

How are you measuring X and Y?

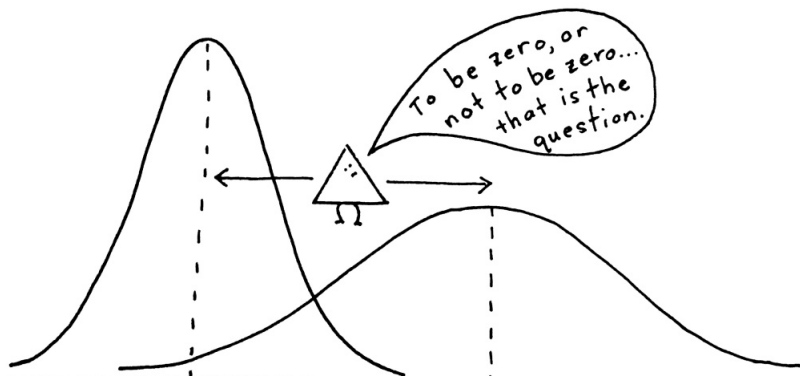
Overview

- Examining Relationships - Hypothesis Testing
- T-tests
- more R fun

Relationships between variables

Today, we'll start to examine the process of being able to assess relationships between two variables.

We'll begin with the simple question of whether there exist statistically significant differences between two groups.

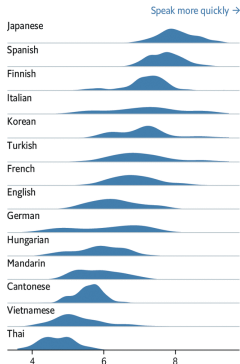


First, Something Interesting

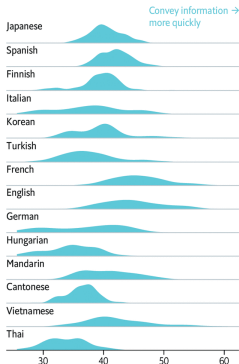
Say no more

Syllable rate and information rate in selected languages

Syllables per second



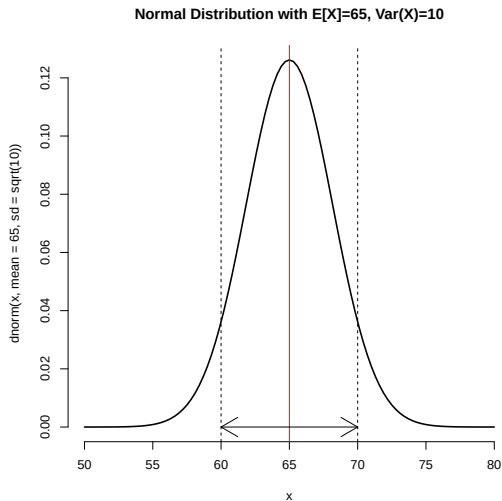
Information rate, bits per second



Source: "Different languages, similar encoding efficiency: Comparable information rates across the human communicative niche" by Christophe Coupé, Yoon Mi Oh, Dan Dediu and François Pellegrino, *Science Advances* (2019)

The Economist

A Random Variable



Dispersion

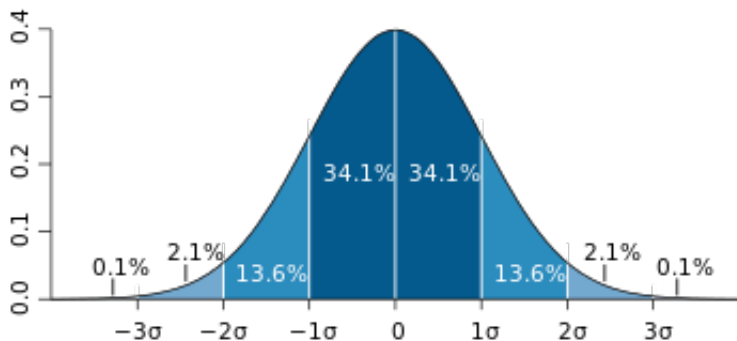
Remember: a random variable is defined by a sample mean and standard deviation.

In other words, we need to think in variables in terms of both their central tendency and their distribution.

$$\text{Variance: } \sigma^2 = \frac{\sum_{i=1}^n (x_i - \mu)^2}{n} \quad \text{Standard Deviation: } \sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \mu)^2}{n}}$$

A Normally Distributed Variable

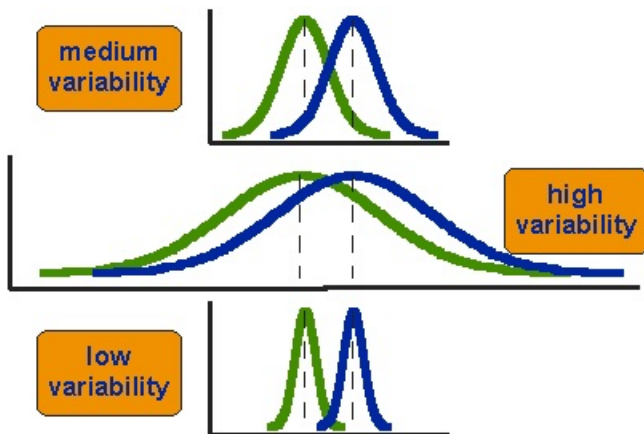
If we know $\bar{x}$ and σ we can calculate percentages of the sample in a given range



Central Limit Theorem

- The mean from a sufficiently large number of independent draws from any distribution will be normally distributed
 - Symmetrical
 - Mean, Median, and Mode at some point
 - asymptotic tails
- the proportion of scores is consistent across standard deviations

Hypothesis Testing - Comparing Random Variables



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We will start by trying to build your intuition by looking at simulations.

Difference in Means

The fundamental quantity of interest today will be the *difference in means* between two groups on some outcome. e.g.

- difference in mean income across two states
- difference in voter turnout among two groups of people (e.g. students vs. employed)
- difference in probability of war in two groups of countries (e.g. autocracies vs. democracies)
- difference in proportion of heads after tossing one coin N_1 time and another coin N_2 times
- difference in treatment vs control groups

Textbook Example: Sheep Experiment

Why did the sheep in one group but not the other group get sick?



Hypotheses

H_0 : Null Hypothesis: A hypothesis of no effect

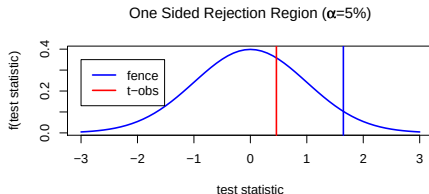
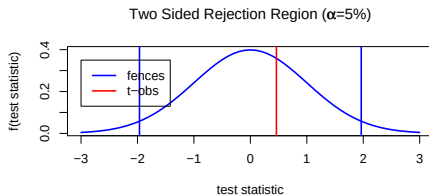
i.e. there is no relationship

the opposite of the alternative hypothesis

H_1 : Alternative Hypothesis: Your actual hypothesis

One-sided vs two-sided tests

- Two sided: Is one group **different**, either BIGGER or SMALLER than the other?
- *Usually we use a two-sided test*



Innocent until proven guilty

- We require a preponderance of the evidence to run against the null hypothesis before we are willing to reject it.
- To put it in statistical terms, we only reject the null hypothesis when the probability of the null hypothesis model producing the observed data is extremely tiny.
- This requires setting an arbitrary threshold, known as the *significance level,* denoted by α , and usually set at $\alpha = 0.05$, sometimes $\alpha = 0.10$, and sometimes $\alpha = 0.01$

Type I vs Type II error

- False positive (Type I error):

Type I vs Type II error

- False positive (Type I error): When we reject a null hypothesis that is actually true - we say there is a relationship when there isn't. The p-value sets the probability of doing this ($0.05=1/20$ th of the time). This is relatively unlikely given a conservative critical value.
- False negative (Type II error):

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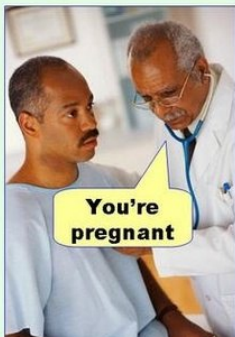
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- False negative (Type II error): When fail to accept the alternative hypothesis, when there is actually a relationship. When is this likely? This could happen if we have a small sample size (the study has low power).

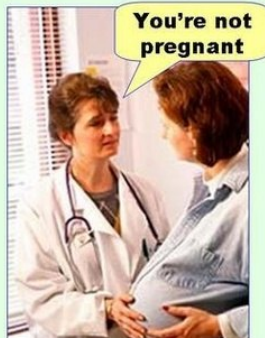
		REALITY	
		NULL HYPOTHESIS	
		TRUE	FALSE
STUDY FINDINGS	TRUE		Type II error (β) 'False negative'
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Type I error
(false positive)



Type II error
(false negative)



Significance

Significance Level - How unlikely a result has to be under the null hypothesis for us to reject the null hypothesis.

We can never rule out that observed result is happening by chance, but we can give a probability to the likelihood that result is occurring by chance

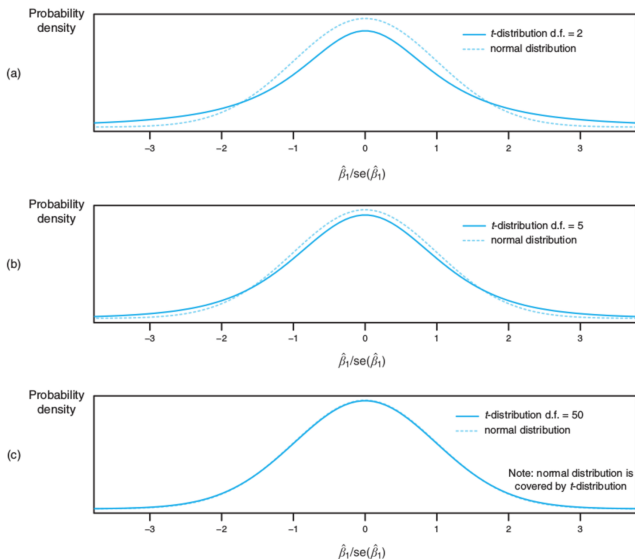
What are the implications of this, especially when we think of most scientific fields establishing by convention a significance level of $\alpha = 0.05$?

The T-Statistic/distribution

- when creating the PDF for β_1 there's one additional bit of uncertainty we need to deal with
- we don't actually know the true variance (or standard error) of β_1 , only the variance of our observed sample $\hat{\beta}_1$
- because this introduces some additional uncertainty, we need to use the t-distribution as opposed to the normal distribution
 - *We will only do this today for illustration purposes*
- as our sample size increases, the t-distribution gets very close to the normal distribution: functionally equivalent when $n > 100$

The t-statistic: $t = \hat{\beta}_1 - \beta^N / \text{se}\hat{\beta}_1$

T-distributions vs. Normal distributions



A T-table

T-Distribution Table (One Tail)

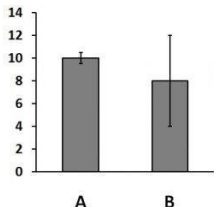
For the T-Distribution Table for Two Tails, [Click Here](#).

DF	$\alpha = 0.1$	0.05	0.025	0.01	0.005	0.001	0.0005
∞	$t_{\alpha} = 1.282$	1.645	1.960	2.326	2.576	3.091	3.291
1	3.078	6.314	12.706	31.821	63.656	318.289	636.578
2	1.886	2.920	4.303	6.965	9.925	22.328	31.600
3	1.638	2.353	3.182	4.541	5.841	10.214	12.924
4	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	1.476	2.015	2.571	3.365	4.032	5.894	6.869
6	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	1.333	1.740	2.110	2.567	2.898	3.646	3.965

A T-test

MR T TEST

"THAT AIN'T SIGNIFICANT, FOOL!"



The t-test gives us a t-statistic that we then compare to our critical value.

In practice today we just look at the p-value our BFF R gives us

p-value

- probability of observing a coefficient as high as we actually observed if the null hypothesis were true
- Probability of committing a type one Error
- Probability happening by chance
- Usually an item is statistically significant if less than .05 (ie less than our critical value)
- We indicate using ***

We then either . . .

- reject the null hypothesis
- fail to reject the null hypothesis

Notice we do NOT "confirm the alternative hypothesis"

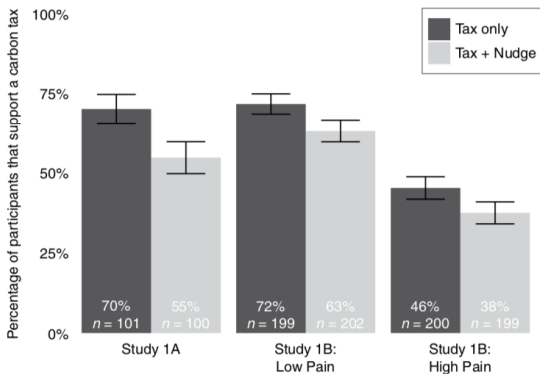


Fig. 1 | Introducing a green energy nudge reduces support for a carbon tax.

Percentage of respondents that support implementing a carbon tax when presented with the tax only or the tax and the nudge (Study 1A, Study 1B: Low Pain) and under conditions in which the carbon tax is framed as more burdensome (Study 1B: High Pain). Error bars show $\pm$ one standard error.

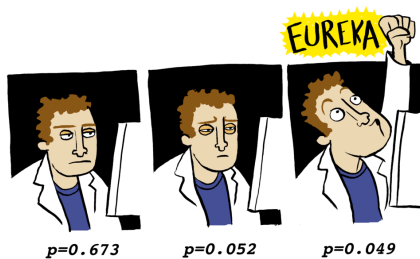
Confidence Intervals

what is the range around our coefficient estimate in which we would expect 95% of our estimates to fall if we continually re-sampled the logic is the same: estimate plus or minus a critical value times the standard error

TABLE 4.6 Calculating Confidence Intervals for Large Samples

Confidence level	Critical value	Confidence interval	Example $\hat{\beta}_1 = 0.41$ and $se(\hat{\beta}_1) = 0.1$
90%	1.64	$\hat{\beta}_1 \pm 1.64 \times se(\hat{\beta}_1)$	$0.41 \pm 1.64 \times 0.1 = 0.246$ to 0.574
95%	1.96	$\hat{\beta}_1 \pm 1.96 \times se(\hat{\beta}_1)$	$0.41 \pm 1.96 \times 0.1 = 0.214$ to 0.606
99%	2.58	$\hat{\beta}_1 \pm 2.58 \times se(\hat{\beta}_1)$	$0.41 \pm 2.58 \times 0.1 = 0.152$ to 0.668

DON'T BE A P-HACKER



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Review of Key Terms

- Hypothesis Testing
- Null Hypothesis
- Alternative Hypothesis
- Degrees of Freedom
- T-test
- Significance level
- Confidence Interval
- Type I error
- Type II error

Problem Sets

General Comments:

- Issues on P-Set #2?
- Is there anything you'd like to review?
- #3 out this weekend
- Will only cover material from today and last week

